

Network Reconnaissance and Protocol Analysis Investigation

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Course: IFT 482

Project Title: Project 1 – Network Reconnaissance and Protocol Analysis

Role: SOC Analyst / Network Forensics Analyst

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Executive Summary

This lab focused on developing practical cybersecurity and network forensics skills through the use of Wireshark and Nmap. The project emphasized packet analysis, host discovery, port scanning, protocol identification, and network reconnaissance techniques commonly used in Security Operations Centers (SOC) and penetration testing environments.

During the investigation, network traffic was captured and analyzed using Wireshark to examine TCP/IP communications, identify protocols, inspect packet structures, and analyze TCP handshakes. Nmap was used to perform host discovery and multiple forms of port scanning, enabling the identification of open ports, running services, and operating systems on networked devices.

The activity provided hands-on experience with traffic analysis, intrusion detection concepts, reconnaissance methodologies, and network visibility practices essential to cybersecurity operations and incident response environments.

Objective

The objective of this project was to:

1. Develop practical experience using Wireshark for packet capture and protocol analysis.
2. Utilize Nmap to conduct host discovery and port scanning activities.
3. Analyze network traffic patterns and identify protocol behavior.
4. Understand reconnaissance techniques used in cybersecurity operations.
5. Gain foundational skills in network forensics and security monitoring.

Tools and Technologies

Security Tools Utilized

Tool	Purpose
Wireshark	Packet capture and protocol analysis
Nmap	Host discovery and port scanning
TCP/IP Protocol Suite	Network communication analysis

Methodology

Wireshark Packet Analysis

The investigation began with configuring Wireshark to monitor the active wireless network interface. Packet captures were collected and analyzed to identify:

- TCP communications
- ARP requests
- ICMP traffic
- HTTP/HTTPS traffic
- MAC address information
- TCP three-way handshakes
- Protocol distribution statistics

Captured packets were filtered using display filters to isolate specific protocols and communication patterns for deeper analysis.

Nmap Reconnaissance

Nmap was used to perform multiple scanning techniques against local systems to identify:

- Active hosts
- Open ports
- Running services
- Operating system fingerprints

- Network exposure points

Various scan types including TCP Connect, SYN, and UDP scans were evaluated to understand their operational behavior and detection characteristics.

Investigation Findings

Phase 1 – Wireshark Traffic Analysis

Network Interface Discovery

The active network interface connected to the local network was identified as:

- **WiFi Interface**

Packet captures confirmed active communication between local systems and external servers.

TCP Three-Way Handshake Analysis

Analysis of captured traffic identified a complete TCP three-way handshake between a client workstation and a remote server.

Component	Value
Client IP Address	192.168.1.18
Server IP Address	162.247.243.29
Client Port	57327
Server Port	443 (HTTPS)

The handshake sequence included:

1. SYN
2. SYN-ACK
3. ACK

This confirmed successful session establishment using TCP over HTTPS.

Packet Inspection Findings

Investigators identified the following during packet analysis:

Observation	Result
Host Value	status.geotrust.com
TCP Traffic Percentage	97.9%
UDP Traffic Percentage	1.2%
UDP Protocol Example	DNS

MAC address analysis revealed:

Device	Manufacturer
Local NIC	Intel
Remote NIC	Netgear

The packet bytes panel confirmed raw hexadecimal and ASCII packet representations useful for forensic analysis.

Phase 2 – ICMP and ARP Analysis

ICMP Traffic Investigation

Wireshark successfully captured ICMP traffic generated through ping requests. The investigation confirmed that:

- ICMP packets were consistently observed during network diagnostics.
- ARP requests were repeatedly generated to resolve MAC addresses.
- ICMP traffic provided insight into host availability and network responsiveness.

ARP Analysis

ARP traffic demonstrated the continuous relationship between IP address resolution and local network communication.

This analysis reinforced understanding of:

- MAC address resolution
- Local subnet communication
- Broadcast domain behavior
- Host discovery mechanisms

Phase 3 – Nmap Reconnaissance and Port Scanning

Host Discovery Results

Nmap scanning activities identified active systems within the local network through:

- ARP requests
- ICMP probes
- TCP connection attempts

Captured traffic showed that Nmap generated multiple packet types during scanning operations.

Open Port Discovery

The scan identified the following open ports:

Open Ports

1080
6543
8888
10001

Scan Statistics

Metric	Result
Total Ports Scanned	996
Open Ports Identified	4

Operating System Identification

Nmap operating system fingerprinting identified the scanned host as:

- **Windows Operating System**

Security Analysis

Port Scanning Techniques Examined

The project evaluated several common reconnaissance scanning techniques:

Scan Type	Purpose
TCP Connect Scan	Full TCP connection establishment
SYN Scan	Stealth half-open scanning
FIN Scan	Firewall and port state analysis
XMAS Scan	Flag manipulation reconnaissance
NULL Scan	Stealth port identification

The investigation included analysis of how these scans appear within packet captures and how defenders can identify them during monitoring operations.

Threat and Detection Concepts

Distributed Slow Scanning

The project explored distributed slow scan methodologies used by attackers to evade detection systems.

Key concepts included:

- Distributed scanning sources

- Randomized timing intervals
- Low-volume reconnaissance
- IDS evasion techniques
- Stealth scanning behavior

TCP Spoofing and Session Hijacking

The investigation also examined advanced attack concepts including:

- IP spoofing
- TCP sequence prediction
- Session hijacking
- ARP poisoning
- Man-in-the-middle attacks

This reinforced the importance of:

- Network segmentation
- Encryption
- Intrusion detection systems
- Secure authentication mechanisms

SOC Analyst Assessment

This project demonstrated the critical role of reconnaissance and packet analysis within cybersecurity operations. Through the use of Wireshark and Nmap, investigators gained visibility into network communications, protocol behavior, host discovery methods, and common scanning techniques used by attackers.

Analysis of packet captures and scanning activity highlighted how network defenders can identify malicious reconnaissance activity through abnormal scanning patterns, repeated connection attempts, protocol anomalies, and unauthorized service discovery. The project further emphasized the importance of packet inspection and traffic analysis in identifying potential indicators of compromise (IOCs).

The investigation also reinforced the significance of understanding TCP/IP communications, ARP behavior, and port scanning methodologies within SOC operations. These skills are essential for threat hunting, intrusion detection, vulnerability assessment, and incident response in enterprise environments.

Recommendations

Security Monitoring Enhancements

1. Deploy intrusion detection systems (IDS) to identify abnormal scanning behavior.
2. Monitor for excessive SYN, FIN, NULL, and XMAS scan activity.
3. Implement network segmentation to reduce lateral reconnaissance opportunities.
4. Enable logging and alerting for repeated host discovery attempts.
5. Utilize packet capture tools for proactive threat hunting.

Defensive Security Practices

- Restrict unnecessary open ports
- Harden exposed services
- Conduct routine vulnerability scanning
- Implement network access controls
- Utilize encrypted communication protocols

Key Skills Demonstrated

Cybersecurity Skills

- Network Traffic Analysis
- Packet Inspection
- Protocol Analysis
- Host Discovery
- Port Scanning
- Threat Detection
- Network Reconnaissance
- TCP/IP Analysis
- Wireshark Packet Filtering
- Nmap Scanning Techniques

Technical Tools

- Wireshark
- Nmap
- TCP/IP Networking
- ICMP Analysis
- ARP Analysis
- Packet Capture Analysis

Conclusion

This project provided practical experience in network reconnaissance, packet analysis, and protocol inspection using industry-standard cybersecurity tools. Through the use of Wireshark and Nmap, critical insights were gained into network behavior, traffic analysis, scanning methodologies, and security monitoring techniques.

The activity strengthened foundational SOC analyst skills by demonstrating how reconnaissance activity appears within network traffic and how defenders can identify suspicious behavior through packet inspection and traffic analysis. The knowledge gained through this investigation directly supports cybersecurity operations including incident response, threat hunting, vulnerability assessment, and digital forensics.